

MSc in Naval Architecture and Ocean Engineering**Modelling and Safety of Maritime Traffic****2026-2027****Assignment****General description:**

The main objective of the assignment is to consolidate the knowledge acquired in Modelling and Safety of Maritime Traffic. By completing the assignment, students are expected to enhance their ability to apply the knowledge gained in the subject, develop critical thinking and reflective skills, and improve their capacity for autonomous learning.

The assignment constitutes a significant 75% of the final course grade. Students are expected to submit a comprehensive report (**Word** and **pdf** files) that provides a detailed explanation of the procedures carried out for each task. Furthermore, each student must include the programming code implemented in the project with appropriate comments.

The project should be submitted solely in digital format on Fénix. For simulation purposes, students may use either MATLAB or Python. The report can be written in either Portuguese or English.

Please note that any instances of plagiarism in the project, or failure to meet the project's main objectives, will result in the group being summoned for an oral discussion.

The assignment comprises five distinct tasks, each contributing different weights to the final evaluation, as outlined below:

1. Turning manoeuvre simulation (2 out of 20 points).
2. Encounter scenario simulation (3 out of 20 points).
3. Assessment of minimum critical distance for give-way start (1 out of 20 points).
4. Minimum safe distance assessment between two vessels (2 out of 20 points).
5. Calculation of collision probability (6 out of 20 points).
6. Assessment of the timely ship collision avoidance (5 out of 20 points)

Additionally, please note that 1 point out of the total 20 is allocated to the overall structure and quality of the report.

Important remarks:

- ! The assignment can be done individually or in groups of two.
- ! The final project submission deadline is **20th October, 23:00**. Please note that there is no requirement to submit a printed version of the project.
- ! Projects should be submitted either through the Fenix platform or, in case of unforeseen issues, via email at mstm@centec.tecnico.ulisboa.pt
- ! Please ensure that all submitted documents are consolidated into a single file, either in ZIP or WinRAR format, and are named as follows: **MSTM_2026_NN_XXXXX_XXXXX**, where NN corresponds to the project number and XXXXX corresponds to the first name of each author in alphabetical order.
- ! Please be aware that the written report must provide sufficient information to comprehensively understand all the procedures undertaken, without the need to refer to the programming code or any external attachments.
- ! For any questions or concerns, please contact the course's official email address: mstm@centec.tecnico.ulisboa.pt

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1 Project description by task

Task 1: Turning manoeuvre simulation

The aim of this task is to conduct a simulation of a turning manoeuvre for Vessel 1 (see Section 2.1) using both the equations of the first-order Nomoto model (Lotovskyi & Teixeira, 2023) and the Manoeuvring Modelling Group (MMG) model (Lotovskyi et al., 2025; Yasukawa & Yoshimura, 2015). It is recommended to begin with an initial heading angle of 0 degrees. The turning circle must be 720 degrees. Students are required to visualise and illustrate the path of the midship point within the turning circle and determine key manoeuvring parameters, such as time to change course 90 degrees, time to change course 180 degrees, advance, transfer, tactical diameter, diameter of turn, speed (steady turn stage), drift (steady turn stage), yaw rate (initial turn stage), yaw rate (steady turn stage). These manoeuvring parameters should be compared with results obtained from official laboratory data (ITTC, 2002; Quadvlieg & Brouwer, 2011; SIMMAN, 2020).

All turning circle trajectories (including the real one) must be presented on a single figure for direct comparison. Please note, the rudder turning speed must be considered.

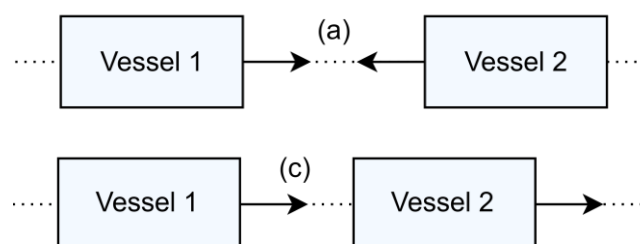
The parameters for the “real” turning manoeuvres can be sourced from the relevant literature (ITTC, 2002; Quadvlieg & Brouwer, 2011; SIMMAN, 2020).

At the end, two MP4 files must be included (i.e., one for each mathematical manoeuvring model), each showing a complete turning circle manoeuvre.

Task 2: Encounter scenario simulation

In this task, students must construct a simulation model based on their assigned Project number, which will depict head-on, crossing, and overtaking situations, using both manoeuvring models. Please comment the obtained results. The converging trajectories of both vessels are illustrated in the figure below. For collision identification, please use the Separating Axis Theorem.

Students are expected to have a clear understanding of which COLREG rule is applicable to each designated encounter scenario (Cockcroft & Lameijer, 2012). These rules should be explicitly identified and explained in the report. Note that, to prevent collision, the vessels will rely solely on rudder adjustments. The simulation model must account for the fact that the vessels initiate the give-way manoeuvre at the minimum safe distance. At the end, the MP4 files must be provided (i.e., one for each encounter scenario and manoeuvring model) showing complete evasive manoeuvres.



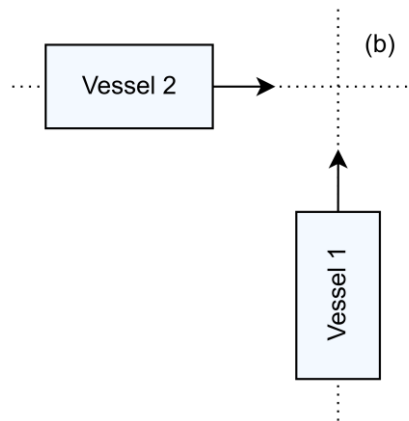


Figure 1. Collision risk assessment of head-on (a), crossing (b), and overtaking (c) situations.

Task 3: Assessment of minimum critical distance for give-way start

The aim of this task is to conduct a comparison and analysis of the minimum safe distance established in Task 2 with the geometric definition of the minimum distance for give-way start as defined by Hashimoto & Okushima (1990). Students are required to provide a comprehensive presentation of all calculations performed in the report.

Task 4: Minimum safe distance assessment between two vessels

The main objectives of this task are: assessment of the minimum safe distance in a specific encounter scenario between two vessels operating in compliance or not with the International Regulations for Preventing Collisions at Sea, COLREGs (depends on the assigned Project number); evaluation of the effects of different manoeuvring models (i.e., Nomoto vs. MMG); calculation of probability of collision using the concept of minimum safe distance (Lotovskyi & Teixeira, 2023).

The minimum safe distance is assessed across two distinct scenarios, defined by the combination of encounter situation and COLREG compliance (Cockcroft & Lameijer, 2012). The procedure is repeated twice: one for each manoeuvring model. Please note, in each scenario, the trajectories of both vessels are simulated using the same manoeuvring model.

To account for stochastic variability and to enhance the robustness of the results, Monte Carlo simulation techniques are incorporated. The recommended number of simulation runs is 10,000, but other numbers may be used if properly justified.

The variables that change between simulation runs are the rudder angle and the approaching speed of both vessels. The variation of rudder angle depends on the simulation scenario and follows one of the following distributions:

$$\delta (^{\circ}) \sim \text{Tri}(1, 15, 35)$$

$$\delta (^{\circ}) \sim \text{U}(-35, 35)$$

$$\delta (^{\circ}) = 0$$

Each ship's speed should follow a uniform distribution. The upper and lower limits for the speeds should be determined by students based on typical service speed variations of similar vessels, which can be found on the MAN official website (MAN, 2021, 2024).

The minimum safe distances should be shown as histograms for each simulation scenario.

Based on the obtained histograms, each group must calculate the probability of collision using the concept of minimum safe distances. The cumulative distribution function (CDF) representing the probability of collision must be plotted overlaid on each corresponding histogram. For more information, please consult Lotovskyi & Teixeira (2023).

Task 5: Calculation of collision probability

The goal of this task is to compute the probability of collision between Vessel 1 and Vessel 2 within the designated canal. All calculations performed should be documented and included in the report. Furthermore, the final result must be thoroughly discussed in light of the probabilities established by Hashimoto & Okushima (1990) and Kuroda et al. (1982) for crossing and head-on; and, Pedersen (1995) and Fujii et al. (1970) for crossing. If inconsistencies are identified in the description of the maritime traffic (Montewka et al., 2010; Silveira et al., 2013), students should clearly indicate which inconsistencies were detected and which solutions were adopted to overcome them.

Task 6: Assessment of the timely ship collision avoidance

The objective of this task is to assess the effect of different ship domain definitions and manoeuvring models on the minimum give-way distance in a **restricted visibility scenario** (more info in Lotovskyi et al. (2026)). For this purpose, four domain-based safety criteria must be implemented (see Figure 2) in a specific relative bearing risk sector (i.e., based on the assigned Project number), see Figure 3.

The distance between ships at which risk of collision exists is analysed through trajectory simulations of two vessels (i.e., Vessel 1 and Vessel 2) under Course alteration diagram of Cockcroft & Lameijer (2012). To ensure sufficient sea room for evasive manoeuvres, two ship domain models are employed: the length-dependent domain (Szlapczynski et al., 2018) and the quaternion domain (Silveira et al., 2022; Wang, 2010), see Figure 4. Ship trajectories in collision scenarios are generated probabilistically using Monte Carlo simulation. The recommended number of simulation runs is 10,000, but other numbers may be used if properly justified. The variables that change between simulation runs are the rudder angle, the approaching speed of both vessels, and the relative bearing angle within the attributed risk sector.

The minimum give-way distances should be shown as histograms for each simulation scenario. Each group must calculate CPA and TCPA. Besides, at least four MP4 files must be included (i.e., one for each domain-based safety criteria using Nomoto or MMG manoeuvring model), each showing a complete timely ship collision avoidance.

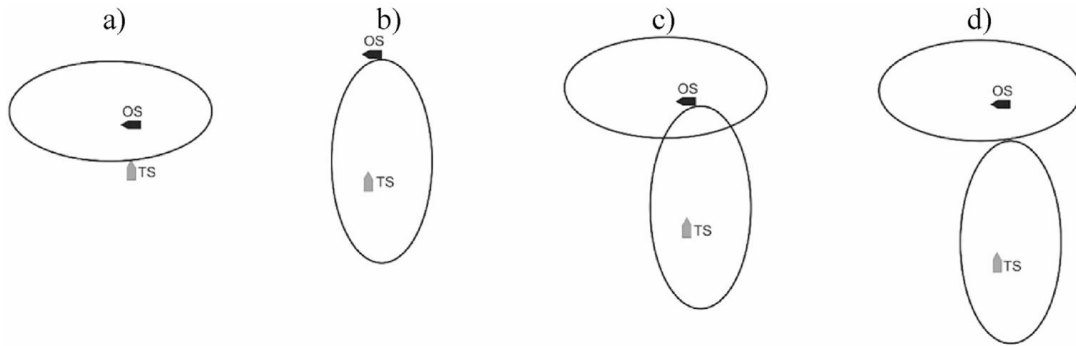


Figure 2. Different domain-based safety criteria: a) OS domain is not violated, b) TS domain is not violated, c) neither OS and TS domain is violated, d) domains do not overlap (Szlapczynski & Szlapczynska, 2017).

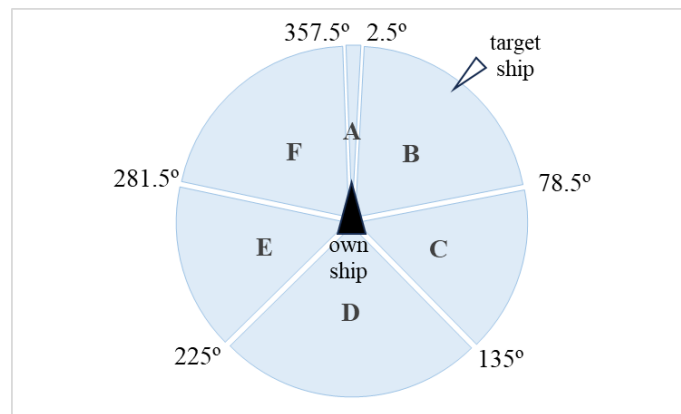


Figure 3. Relative bearing risk sectors (Silveira et al., 2021).

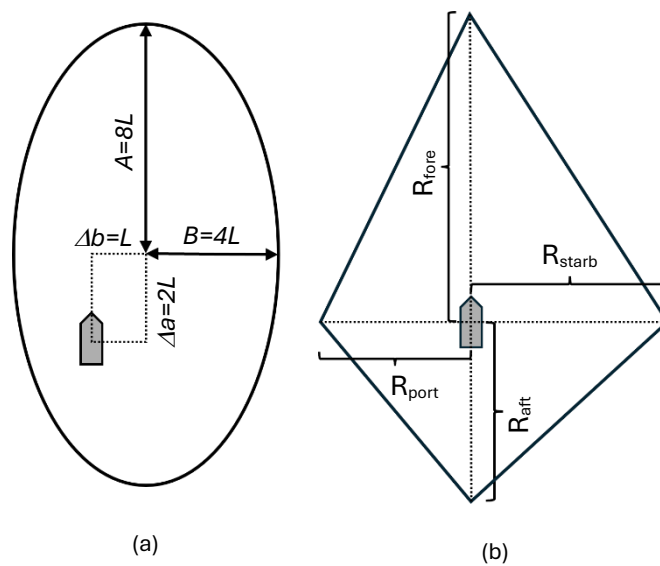


Figure 4. General dimensions and geometric characteristics of the two ship domain configurations used in the case study: (a) Length-dependent ship domain; (b) Quaternion ship domain.

2 Project data

2.1 General information

In the following table each project is summarised with its associated tasks:

Project	Vessels		Minimum safe distance			Probability of collision in Head-on			Probability of collision in Crossing (TSS Off Cape Roca vs. Lisbon-Madeira)		Relative bearing risk sector
	Vessel 1	Vessel 2	Collision model	Scenario 1	Scenario 2	Shipping canal	Direction of Vessel 1	Direction of Vessel 2	Direction of Vessel 1	Direction of Vessel 2	
1	KVLCC2	ESSO Osaka	Head-on	COLREG-COLREG	Blind-Blind	Gulf of Finland	East	West	Lisbon	South	A
2	ESSO Osaka	KCS	Crossing	COLREG-COLREG	Blind-Blind	Lisbon-Madeira	Madeira	Lisbon	Madeira	North	B
3	KCS	KVLCC2	Overtaking	COLREG-COLREG	Blind-Blind	Gulf of Finland	West	East	North	Madeira	C
4	KVLCC2	KCS	Head-on	COLREG-Blind	Blind-COLREG	Lisbon-Madeira	Lisbon	Madeira	South	Lisbon	D
5	ESSO Osaka	KVLCC2	Crossing	COLREG-Blind	Blind-COLREG	Lisbon-Madeira	Lisbon	Madeira	Madeira	North	E
6	KCS	Esso Osaka	Overtaking	COLREG-Blind	Blind-COLREG	Gulf of Finland	West	East	Lisbon	South	F
7	KVLCC2	ESSO Osaka	Crossing	COLREG-COLREG	Blind-Blind	Lisbon-Madeira	Lisbon	Madeira	Madeira	North	B
8	ESSO Osaka	KCS	Overtaking	COLREG-COLREG	Blind-Blind	Gulf of Finland	West	East	Lisbon	North	C
9	KCS	KVLCC2	Head-on	COLREG-COLREG	Blind-Blind	Lisbon-Madeira	Lisbon	Madeira	South	Lisbon	D
10	KVLCC2	KCS	Crossing	COLREG-Blind	Blind-COLREG	Gulf of Finland	West	East	North	Madeira	E
11	ESSO Osaka	KVLCC2	Overtaking	COLREG-Blind	Blind-COLREG	Gulf of Finland	East	West	Lisbon	South	F
12	KCS	Esso Osaka	Head-on	COLREG-Blind	Blind-COLREG	Lisbon-Madeira	Madeira	Lisbon	Madeira	South	A
13	KVLCC2	ESSO Osaka	Overtaking	COLREG-COLREG	Blind-Blind	Gulf of Finland	West	East	Lisbon	North	C
14	ESSO Osaka	KCS	Head-on	COLREG-COLREG	Blind-Blind	Lisbon-Madeira	Lisbon	Madeira	Madeira	South	D
15	KCS	KVLCC2	Crossing	COLREG-COLREG	Blind-Blind	Gulf of Finland	East	West	North	Lisbon	E
16	KVLCC2	KCS	Overtaking	COLREG-Blind	Blind-COLREG	Lisbon-Madeira	Madeira	Lisbon	South	Madeira	F
17	ESSO Osaka	KVLCC2	Head-on	COLREG-Blind	Blind-COLREG	Lisbon-Madeira	Madeira	Lisbon	Madeira	South	A
18	KCS	Esso Osaka	Crossing	COLREG-Blind	Blind-COLREG	Gulf of Finland	East	West	Lisbon	North	B
19	KVLCC2	ESSO Osaka	Head-on	COLREG-COLREG	Blind-COLREG	Lisbon-Madeira	Madeira	Lisbon	Madeira	South	D
20	ESSO Osaka	KCS	Crossing	COLREG-COLREG	Blind-COLREG	Gulf of Finland	East	West	Lisbon	South	E
21	KCS	KVLCC2	Overtaking	COLREG-COLREG	Blind-COLREG	Lisbon-Madeira	Madeira	Lisbon	South	Madeira	F
22	KVLCC2	KCS	Head-on	COLREG-Blind	Blind-Blind	Gulf of Finland	East	West	North	Lisbon	A
23	ESSO Osaka	KVLCC2	Crossing	COLREG-Blind	Blind-Blind	Gulf of Finland	West	East	Lisbon	North	B
24	KCS	Esso Osaka	Overtaking	COLREG-Blind	Blind-Blind	Lisbon-Madeira	Lisbon	Madeira	Madeira	North	C

2.2 Benchmark vessels

For the purposes of this project, three reference vessels are to be considered: KVLCC2, KCS, and ESSO Osaka.

KVLCC2 and **KCS** are benchmark ship models developed by the Korea Research Institute of Ships and Ocean Engineering (KRISO): the KRISO Very Large Crude Carrier 2 (KVLCC2) and the KRISO Container Ship (KCS).

KVLCC2 is a modern tanker hull form with a bulbous bow and transom stern. KCS is a modern container ship with a bulbous bow and stern. Both were conceived in 1997. These two hull forms are among four ship models selected by the 24th ITTC Manoeuvring Committee to establish a new benchmark dataset (ITTC, 2021). Although neither vessel exists at full scale, both accurately represent modern hull designs for their respective ship types.

In the simulations, the initial approach speed is set to *15.5 knots for KVLCC2* and *24 knots for KCS*. The rudder steering rate is *2.33 degrees per second*, and the propeller revolution is maintained at a constant rate corresponding to the initial approach speed. The KVLCC2 manoeuvre is assessed in a starboard turning circle, while the KCS manoeuvre is assessed in a portside turning circle.

ESSO Osaka was a tanker vessel constructed by Hitachi Zosen in Japan in 1973. The vessel operated until 1985, when she was scrapped at Kaohsiung. According to ITTC (2002) the main advantages of ESSO Osaka trials data are an unusually extensive set of trials of the Esso Osaka at full load draught, captive model tests or free running model tests of the ESSO Osaka had been carried out by at least 20 different laboratories, the drawings of the ESSO Osaka hull, propeller and rudder had been available by EXXON. Besides, as the primary disadvantages, ITTC (2002) highlighted the following factors: the relative old hull form of a ship launched in 1973, the unavailability of good quality resistance or propulsion data. However, the shortcomings of the ESSO Osaka data were considered minor when compared to with their advantages. In the simulations, the initial approach speed is set to *10 knots for ESSO Osaka*.

Table 1. Principal characteristics of KVLCC2 (Lotovskyi et al., 2025).

Variables	Data	Variables	Data	Variables	Data	Variables	Data
L_{PP} (m)	320	a_0	0.2931	Λ	1.826	l'_R	-0.710
B (m)	58	a_1	-0.2753	t_R	0.387	a_H	0.312
d (m)	20.8	a_2	0.1385	ε	1.09	μ'_{11}	0.022
U (knots)	15.5	n_p	1.2413	w_{R_0}	0.3192	μ'_{22}	0.223
m (t)	320437	w_{P_0}	0.39202	κ	0.5	μ'_{26}	0.0082
C_B	0.81	t_P	0.22	γ_R^+	0.395	μ'_{66}	0.011
x_G (m)	11.1	A_R (m ²)	136.7	γ_R^-	0.640	R'_0	0.022
D_P (m)	9.86	H_R (m)	15.8	x'_H	-0.40052		

Table 2. Hydrodynamic derivatives of KVLCC2 (Lotovskyi et al., 2025).

Variables	Data	Variables	Data	Variables	Data	Variables	Data
X'_{vv}	-0.04	Y'_v	-0.315	Y'_{vrr}	-0.391	N'_{vvv}	-0.03
X'_{vr}	0.002	Y'_r	0.083	Y'_{rrr}	0.008	N'_{vvr}	-0.294
X'_{rr}	0.011	Y'_{vvv}	-1.607	N'_v	-0.137	N'_{vrr}	0.055
X'_{vvvv}	0.771	Y'_{vvr}	0.379	N'_r	-0.049	N'_{rrr}	-0.013

Table 3. Principal characteristics of KCS (Lotovskyi et al., 2025).

Variables	Data	Variables	Data	Variables	Data	Variables	Data
L_{PP} (m)	230	a_0	0.472	Λ	1.80	l'_R	-0.755
B (m)	32.2	a_1	-0.3073	t_R	0.258	a_H	0.361
d (m)	10.8	a_2	-0.1354	ε	0.956	μ'_{11}	0.0057
U (knots)	24	n_P	1.7322	w_{R_0}	0.315	μ'_{22}	0.188
m (t)	53330	w_{P_0}	0.355	κ	0.633	μ'_{26}	-0.00255
C_B	0.651	t_P	0.207	γ_R^+	0.338	μ'_{66}	0.00695
x_G (m)	-3.4	A_R (m ²)	54.45	γ_R^-	0.492	R'_0	0.0167
D_P (m)	7.9	H_R (m)	9.9	x'_H	-0.436		

Table 4. Hydrodynamic derivatives of KCS (Lotovskyi et al., 2025).

Variables	Data	Variables	Data	Variables	Data	Variables	Data
X'_{vv}	-0.0549	Y'_v	-0.2252	Y'_{vrr}	-0.8341	N'_{vvv}	-0.1751
X'_{vr}	-0.0797	Y'_r	0.04705	Y'_{rrr}	-0.005	N'_{vvr}	-0.6167
X'_{rr}	-0.0120	Y'_{vvv}	-1.7179	N'_v	-0.1111	N'_{vrr}	-0.0512
X'_{vvvv}	-0.0417	Y'_{vvr}	-0.4832	N'_r	-0.0465	N'_{rrr}	-0.0387

Table 5. Principal characteristics of ESSO Osaka (ITTC, 2002).

Variables	Data	Variables	Data	Variables	Data	Variables	Data
L_{PP} (m)	325	a_0	0.3488	Λ	1.601	l'_R	-0.82
B (m)	53	a_1	-0.3376	t_R	0.2173	a_H	0.398
d (m)	21.8	a_2	-0.1395	ε	1.42	μ'_{11}	0.02
U (knots)	10.0	n_P	0.8383	w_{R_0}	0.3512	μ'_{22}	0.2224
m (t)	314,410	w_{P_0}	0.614	κ	0.288	μ'_{26}	0.0098
C_B	0.830	t_P	0.2	γ_R^+	0.4080	μ'_{66}	0.00821
x_G (m)	10.3	A_R (m ²)	119,817	γ_R^-	0.4080	R'_0	0.0117
D_P (m)	9.1	H_R (m)	13.85	x'_H	-0.442		

Table 6. Hydrodynamic derivatives of ESSO Osaka (ITTC, 2002).

Variables	Data	Variables	Data	Variables	Data	Variables	Data
X'_{vv}	-0.00329	Y'_v	-0.3831	Y'_{vrr}	-0.25589	N'_{vvv}	0.053257
X'_{vr}	-0.038784	Y'_r	0.082145	Y'_{rrr}	-0.01119	N'_{vvr}	-0.29699
X'_{rr}	0.000034	Y'_{vvv}	-1.05375	N'_v	-0.14716	N'_{vrr}	0.023637
X'_{vvvv}	0.277875	Y'_{vvr}	0.59837	N'_r	-0.04836	N'_{rrr}	-0.01835

2.3 Shipping canal data

2.3.1 TSS off Cape Roca and Lisbon-Madeira route

The maritime areas under national jurisdiction off the continental coast of Portugal are crossed by busy commercial routes connecting northern Europe and the Mediterranean Sea. A significant part of the maritime traffic coming from the Far East, particularly China, passes through the Suez Canal, crosses the Mediterranean Sea to the strait of Gibraltar and heads towards northern Europe through Portuguese waters. Here ships meet vessels bound to and leaving from national ports, fishing vessels, pleasure craft, traffic from the Mediterranean to North America, from Africa to northern Europe, and from northern Europe to South America, creating a complex network of routes.

There are two TSSs off the Portuguese coast, off Cape Roca and off Cape San Vicente. There is also one area to be avoided, protecting the environmentally sensitive Berlengas archipelago. Both TSSs had, at the time when the data was collected, four separated traffic lanes, four separation zones and one Inshore Traffic Zone.

The four lanes in each TSS are:

- Northbound non-dangerous cargo traffic lane
- Northbound dangerous cargo traffic lane
- Southbound non-dangerous cargo traffic lane
- Southbound dangerous cargo traffic lane

The number of ships and average speeds on the traffic lanes of the TSS off Cape Roca is shown in the first table:

Table 7. Number of ships and average speeds, off Cape Roca TSS traffic lanes, 32-day time period.

Average speed (knots)/Number of Ships			
Northbound lanes		Southbound lanes	
Dangerous	Non dangerous	Dangerous	Non dangerous
13-13/292	13-75/1283	13-95/262	14-86/1237

For the purpose of the Assignment, the route between the port of Lisbon and the archipelago of Madeira that crosses the TSS off Cape Roca is called Lisbon-Madeira route. The route is delimited by points P1 (38° 30.86'N/009°30.96'W) and P2 (38° 01.02'N/010° 09.05'W). Please consider that the width of the route is 2 nm (see Figure 5).

The results show that the route Lisboa to Madeira was used 26 times and that the route Madeira to Lisboa was used 12 times. It was also registered that this part of the route was crossed 1414 times from the NW to the SE, and 1521 times from the SE to the NW.

The second table shows some information about traffic in the TSS off Cape Roca. Average speeds were calculated for ships with speed over 1 knot to discard vessels at anchor or drifting.

For more information, please consult:

- ! Silveira, P.A.M., Teixeira, A.P., & Guedes Soares, C. 2013. Use of AIS Data to Characterise Marine Traffic Patterns and Ship Collision Risk off the Coast of Portugal. *The Journal of Navigation*, 66: 879–898.
- ! <https://map.openseamap.org/>

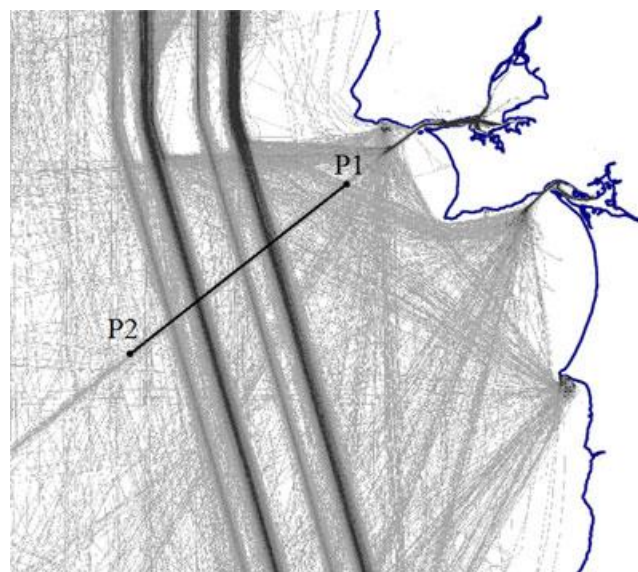


Figure 5. Traffic density and route between point 1 and point 2.

Table 8. General traffic information for the studied area.

Average number of ships:	139·92
Average speed of ships:	12·81 knots
Average number of passenger ships:	1·96
Average speed of passenger ships ($v \geq 1$ knot):	18·41 knots
Average length of passenger ships:	233·17 metres
Average number of cargo ships:	95·94
Average speed of cargo ships ($v \geq 1$ knot):	13 knots
Average length of cargo ships:	146·83 metres
Average number of tankers:	27·57
Average speed of tankers ($v \geq 1$ knot):	12·99 knots
Average length of tankers:	161·73 metres
Average number of other ships:	11·06
Average speed of other ships ($v \geq 1$ knot):	10·11 knots
Average length of other ships:	88·1 metres

2.3.2 Gulf of Finland

The analysed area of the Gulf of Finland corresponds to the junction of two main waterways: one leads N-S and the other E-W. The N-S stream consists mainly of passenger vessels cruising between Helsinki and Tallinn, whereas the E-W stream consists of cargo vessels bound for and from harbours located in the Gulf of Finland.

The vessel traffic profiles over the area under analysis are described using data derived from the AIS transmissions recorded in March and July 2006. The traffic recorded in March is considered the winter profile of traffic, whereas the traffic registered in July is a summer profile of marine traffic in the Gulf of Finland.

The mode and maximum values of length, breadth and draught in each class of vessel are presented in the first table. The mode value denotes the frequently recorded values for a given class of vessel, so they may also be called typical parameters of vessels operating in the Gulf of Finland in 2006. The total number of vessels registered at gates 1 and 2 is presented in the second table. There are no significant differences in the number of vessels passing gate 1 during each season (i.e., winter, summer). However, such differences exist for gate 2, where the number of vessels in summer is marked with an asterisk and for winter with double asterisk. The main reason for the difference is that during wintertime the operations of high-speed craft between Helsinki and Tallinn were suspended and, therefore, the number of passenger vessels on the above route was reduced.

For more information, please consult:

- ! Montewka, J., Hinz, T., Kujala, P., & Matusiak, J. 2010. Probability modelling of vessel collisions. *Reliability Engineering & System Safety*, 95(5): 573–589.
- ! <https://map.openseamap.org/>

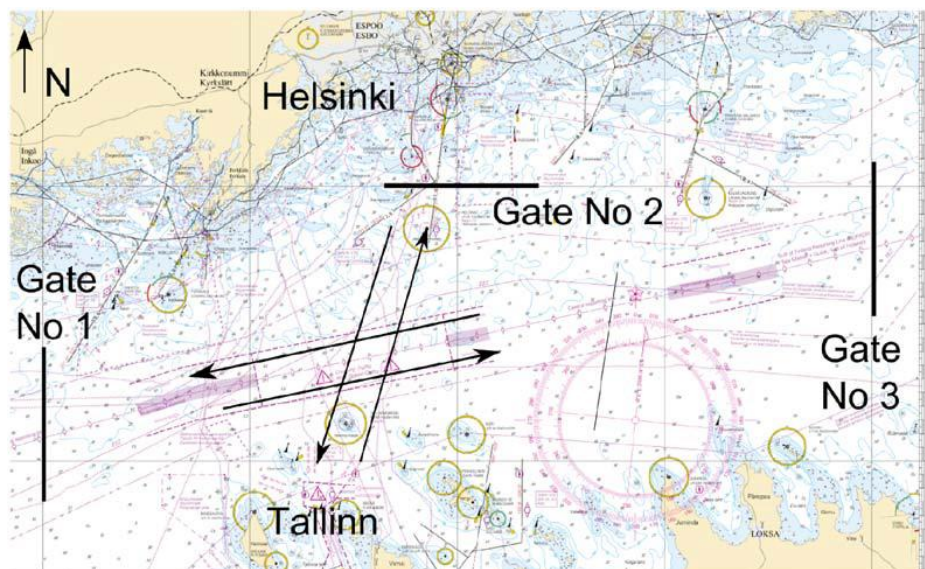


Figure 6. Waterway junction of the Gulf of Finland with main traffic flows marked.

Types and sizes of typical and maximal vessels operating in the Gulf of Finland in the year 2006.

Vessel type	L_mode	L_max	B_mode	B_max	T_mode	T_max
Ro-ro	162.0	195.0	20.6	28.1	6.7	10.0
Tankers	239.0	320.0	27.3	58.0	11.2	22.0
Container carriers	124.5	192.7	22.5	25.4	8.7	10.5
Passenger ships	171.3	266.2	28.7	36.0	5.0	8.5
General cargo ships	85.0	173.5	12.5	27.6	5.3	10.9

Number of vessels passing gate 1 (E–W traffic) and gate 2 (N–S traffic), registered in March (**) and July (*) 2006.

Vessel type	Gate number 1		Gate number 2	
	East	West	North	South
Ro-ro	397	400		
Tankers	618	607		
Container carrier	470	464		
Passenger ship	505	502	1140* 488**	1140* 488**
General cargo ships	895	929		
Bulk carriers	153	145		
Total	3038	3047		

2.4 Quaternion ship domain

	Length:100–200 m Cargo ships (Type: 70–79)	Length: 100–200 m Tankers (Type: 80–89)	Length: 200–300 m Cargo ships (Type: 70–79)	Length:200–300 m Tankers (Type: 80–89)
Differential Evolution/Dual Annealing algorithms				
k	1.789	1.841	2.007	2.118
R_{fore}	0.451	0.719	0.959	1.146
R_{aft}	0.488	0.657	0.936	0.916
R_{sb}	0.206	0.357	0.291	0.508
R_{port}	0.193	0.309	0.307	0.519